LOST AND FOUND RFP

Subject: Lost and Found Metrics pseudocode

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Date: 24 February 2026

Reference: MET-2

Version: 1.0.0

1. System uptime must be ≥ 99%

Pseudocode:

```
SET AppURL = "https://api.lostandfoundapp.com/health"
SET CheckIntervalSeconds = 60
SET TotalChecks = 0
SET SuccessfulChecks = 0

// Main Monitoring Loop
WHILE SystemRunning:
  SET StartTime = GetCurrentTime()
  TotalChecks++

  // Perform Health Check
  TRY:
    Response = HTTP_GET(AppURL)
    IF Response.StatusCode == 200:
      SuccessfulChecks++
      Log("App is UP")
    ELSE:
      Log("App is DOWN - Status: " + Response.StatusCode)
      TriggerAlert("Lost & Found App is Down!")
  CATCH Error:
```

```
        Log("App is UNREACHABLE - Error: " + Error)
        TriggerAlert("Lost & Found App is Unreachable!")

    // Calculate and Display Uptime
    UptimePercentage = (SuccessfulChecks / TotalChecks) * 100
    Display("Current Uptime: " + UptimePercentage + "%")

    // Wait for next check
    Wait(CheckIntervalSeconds)

FUNCTION TriggerAlert(message):
    // Logic to send Email/SMS/Slack alert to admin
    SendNotification(AdminEmail, message)
END FUNCTION
```

2. Item recovery rate: Rate at which lost item is recovered/returned
   Pseudocode:

```
FUNCTION CalculateItemRecoveryRate(time_period)
        INTEGER totalLostReports
        INTEGER recoveredItems
        FLOAT recoveryRate

        totalLostReports = 0
        recoveredItems = 0

        totalLostReports = Items.query(SELECT COUNT(*)
                                       WHERE status = 'Lost'
                                       AND date_reported WITHIN time_period)

        recoveredItems = Items.query(SELECT COUNT(*)
                                       WHERE status = 'recovered'
                                       AND date_reported WITHIN time_period)
```

```
        IF totalLostReports == 0
            return = 0
        ELSE
            recoveryRate = (recoveredItems / totalLostReports) * 100
            RETURN recoveryRate
END FUNCTION
```

3. Search query response time

   Pseudocode:

```
FUNCTION sgetQuerySearchResponseTime(searchQuery):
  // 1. Record start time
  startTime = getCurrentTimestamp() // in milliseconds

  // 2. Perform the database/search query
  results = Database.find(query = searchQuery)

  // 3. Record end time
  endTime = getCurrentTimestamp()

  // 4. Calculate duration (latency)
  responseTime = endTime – startTime

RETURN responseTime
END FUNCTION
```